

MODULE 2: TURBINES AND IC ENGINES

A. Steam turbines: Classification, Principle of operation of Impulse and reaction turbines, Delaval's turbine, Parson's turbine. *(No compounding of turbines)*. **Gas turbines:** Classification, Working principles and Operations of Open cycle and closed cycle gas turbines. **Water turbines;** Classification, Principles and operations of Pelton wheel, Francis turbine and Kaplan turbine

B. Internal Combustion Engines

Classification, I.C. Engines parts, 2 Stroke and 4 stroke Petrol engines, 4 stroke diesel engines. P-V diagrams of Otto and Diesel cycles. Problems on indicated power, brake power, indicated thermal efficiency, brake thermal efficiency, mechanical efficiency, and specific fuel consumption, *[numericals on IC Engines]*.

TURBINES

Prime Mover: Prime Mover is a device which converts the available natural source of energy into mechanical energy/work

Ex: Steam Turbines, Gas turbine, Water turbines, Petrol engines, Diesel engines

The work obtained from the prime mover is used to drive other machines

Ex:

- To drive the electrical generators
- To drive the pumps
- To drive the compressors

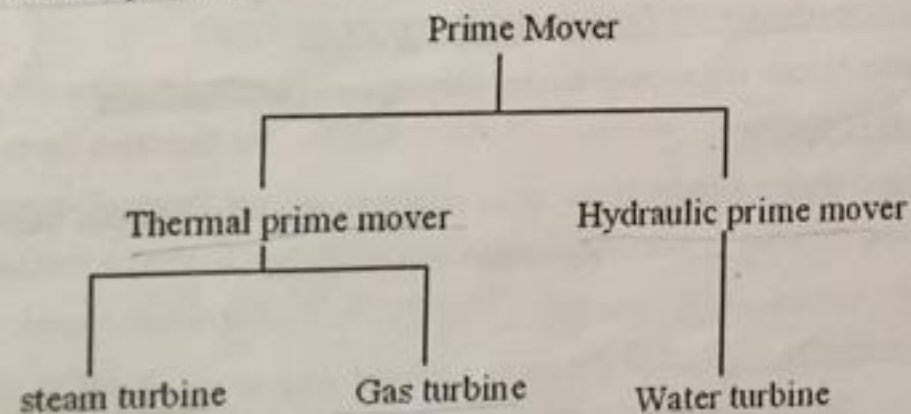


Fig: Classification of Prime movers

TURBINE: Turbine is a prime mover having curved blades on its periphery. Generally turbines are used to drive the electrical generators to produce the electricity

- The turbine consists of a series of curved blades fixed on the circumference of a single wheel called rotor
- Rotor is connected to a shaft

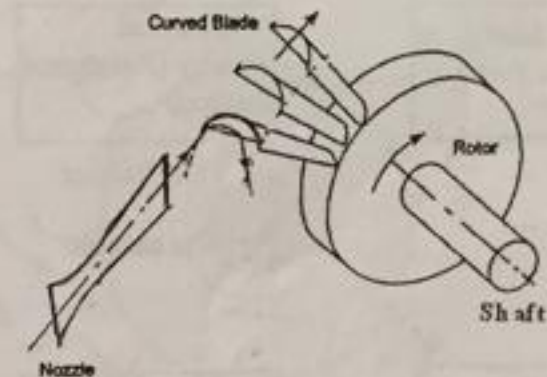


Fig: Turbine

Types of Turbines

1. Steam Turbines
2. Gas Turbines (Combustion Turbines)
3. Water (Hydraulic) Turbines

Steam Turbine: A steam turbine is a thermal prime mover that converts the heat energy of steam into Mechanical energy in the form of rotation of shaft. The mechanical energy is converted into Electrical energy using an electrical generator

Gas turbine: A gas turbine is a thermal prime mover that converts the heat energy of hot air or burnt gases into Mechanical energy in the form of rotation of shaft. The mechanical energy is converted into Electrical energy using an electrical generator

Water turbine: A water turbine is a hydraulic prime mover that converts the hydraulic energy into Mechanical energy in the form of rotation of shaft. The mechanical energy is converted into Electrical energy using an electrical generator

TYPES OF STEAM TURBINE ARE

1. IMPULSE TURBINE.
2. REACTION TURBINE.

1. IMPULSE STEAM TURBINE OR DE-LAVAL TURBINE:

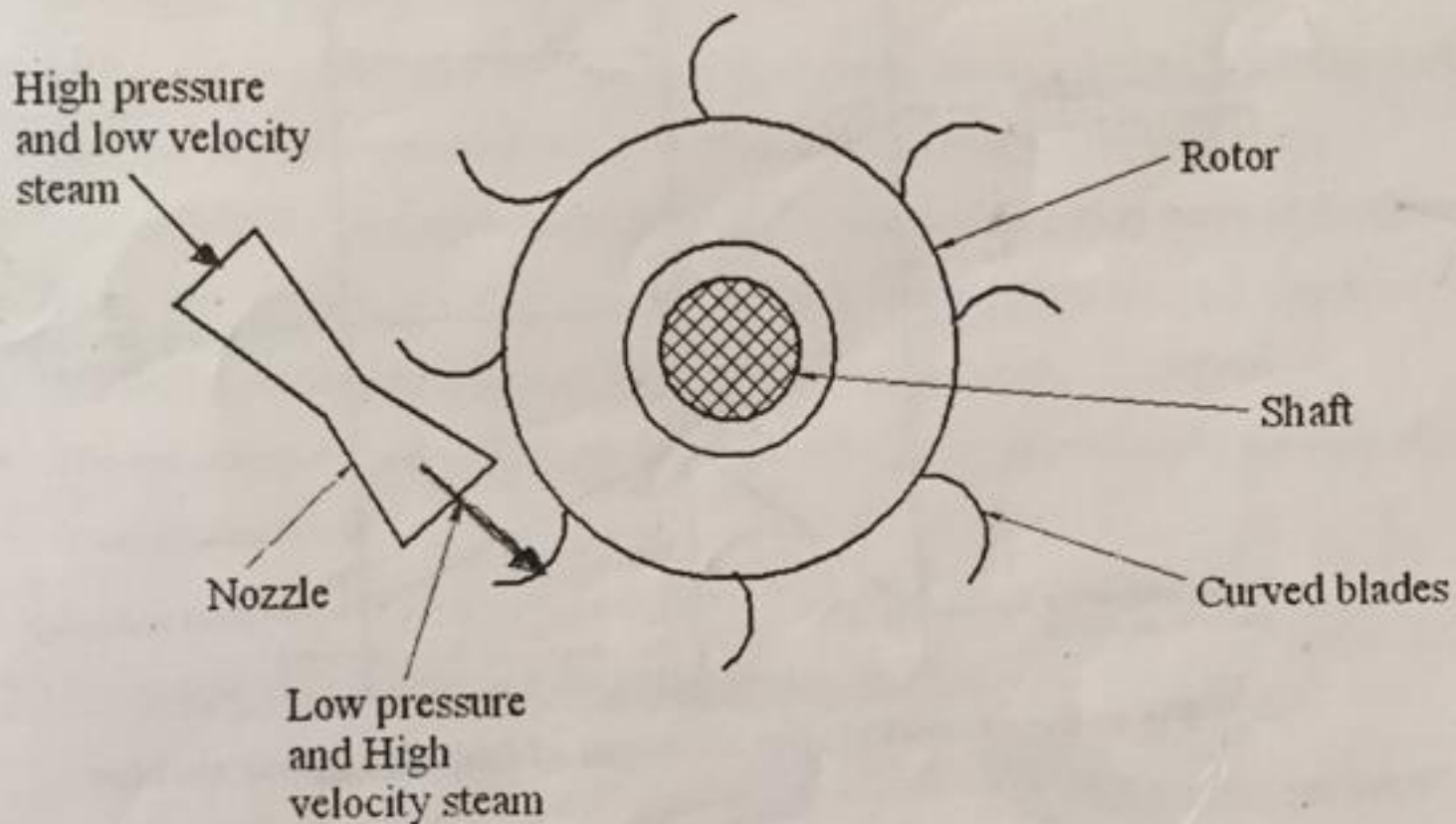


Fig a: Impulse Steam Turbine

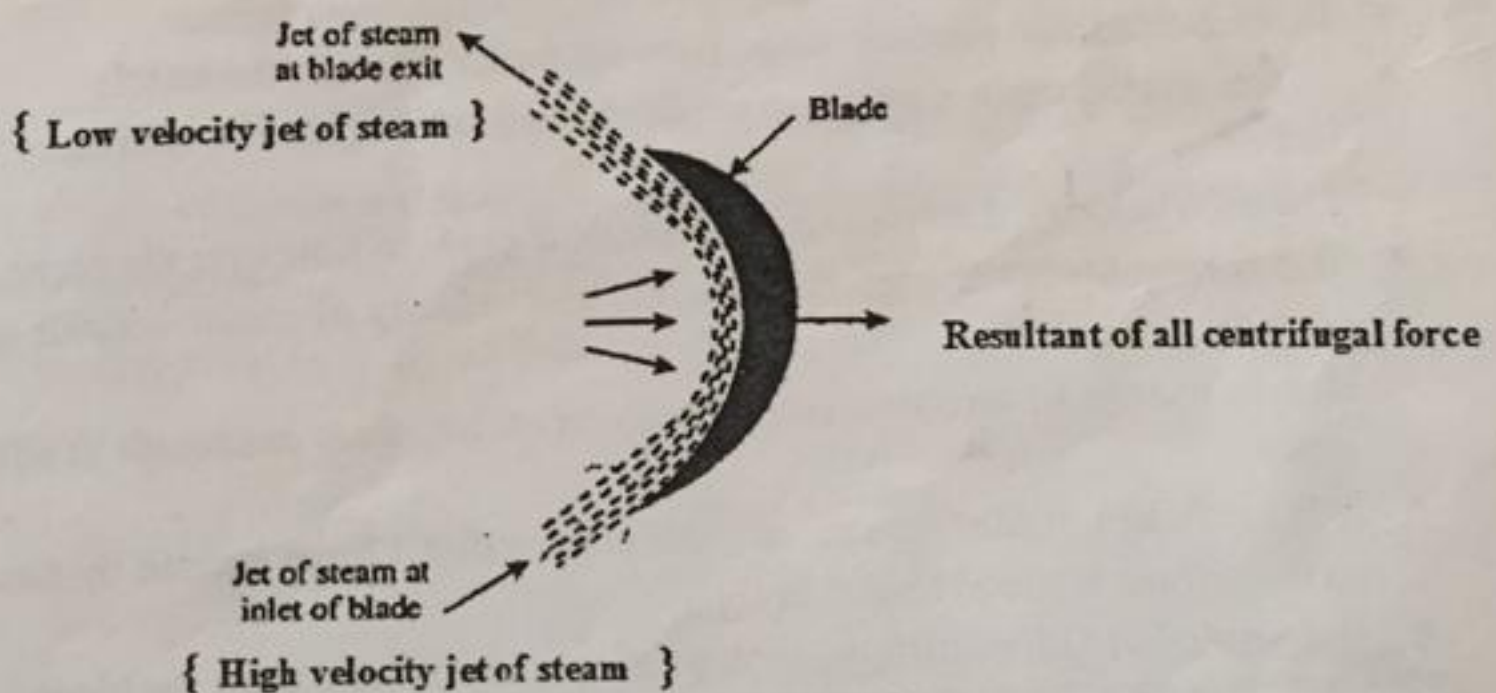


Fig b: Forces acting on the blade

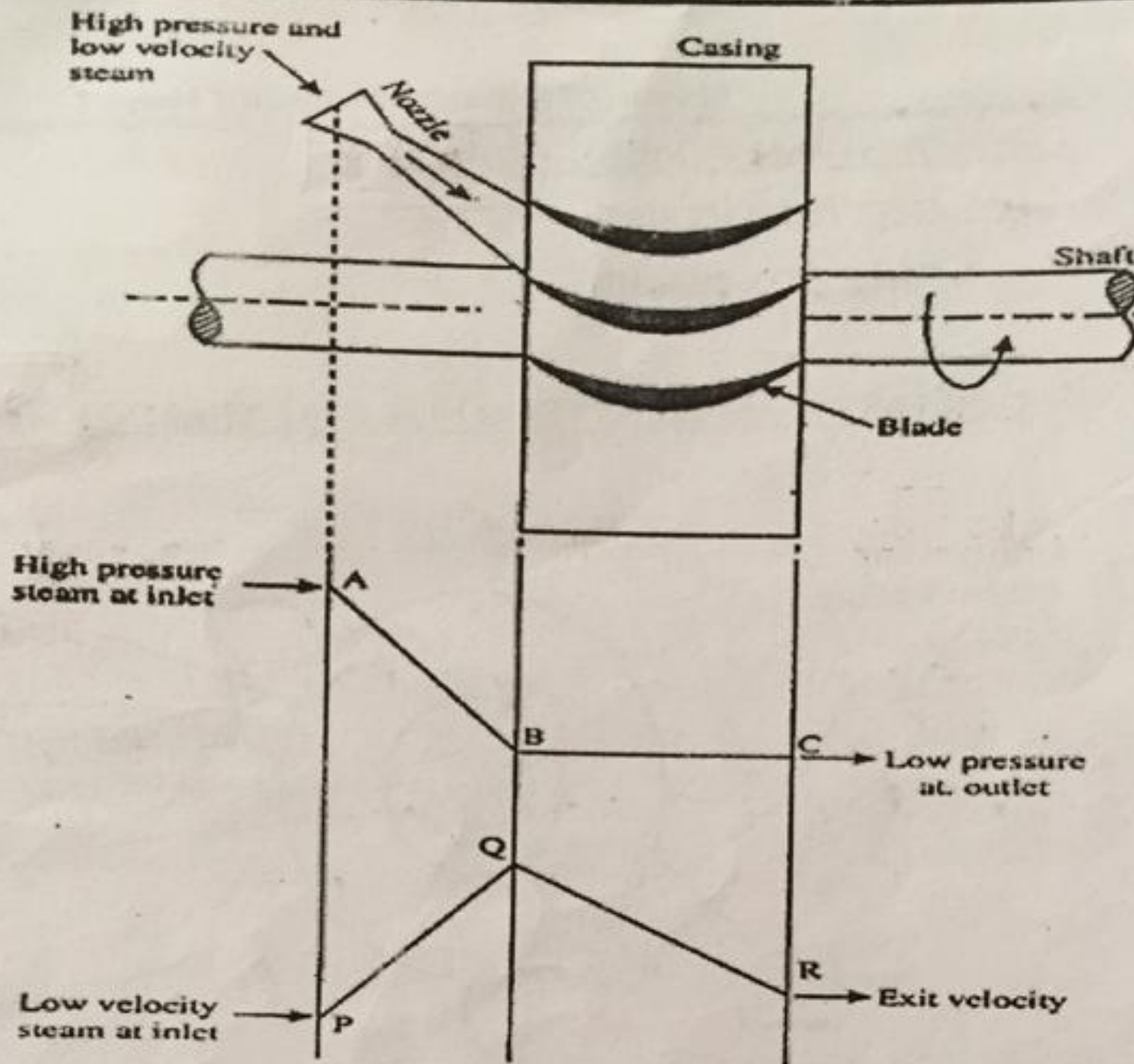


Fig c: Pressure-Velocity diagram of Impulse steam turbine

Working principle:

- Fig a shows Impulse steam turbine
- Fig b shows Forces acting on the blade
- Fig c shows Pressure-Velocity diagram of Impulse steam turbine
- In operation, high pressure steam from the boiler enters to the nozzle
- In the nozzle steam expands **completely** and hence pressure decreases and velocity increases
- The high velocity steam leaving the nozzle is made to flow over the curved blades
- The curved blades change the direction and velocity of steam without changing the pressure
- Due to change in direction and velocity of steam, there is change in Momentum of steam
- Due to change in Momentum, impulse or centrifugal force exerted by steam particles on the turbine blade as shown in fig a
- The resultant of all centrifugal forces acting on the blade causes the blade to forward

WORKING PRINCIPLE OF REACTION STEAM TURBINE

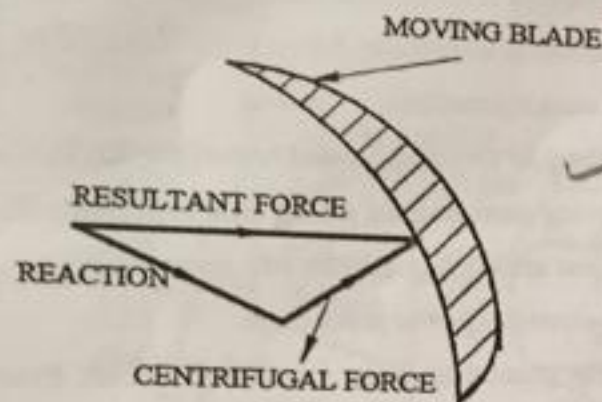


Fig b: Forces acting on the Moving blade of reaction turbine

P-V Diagram of Reaction Steam Turbine

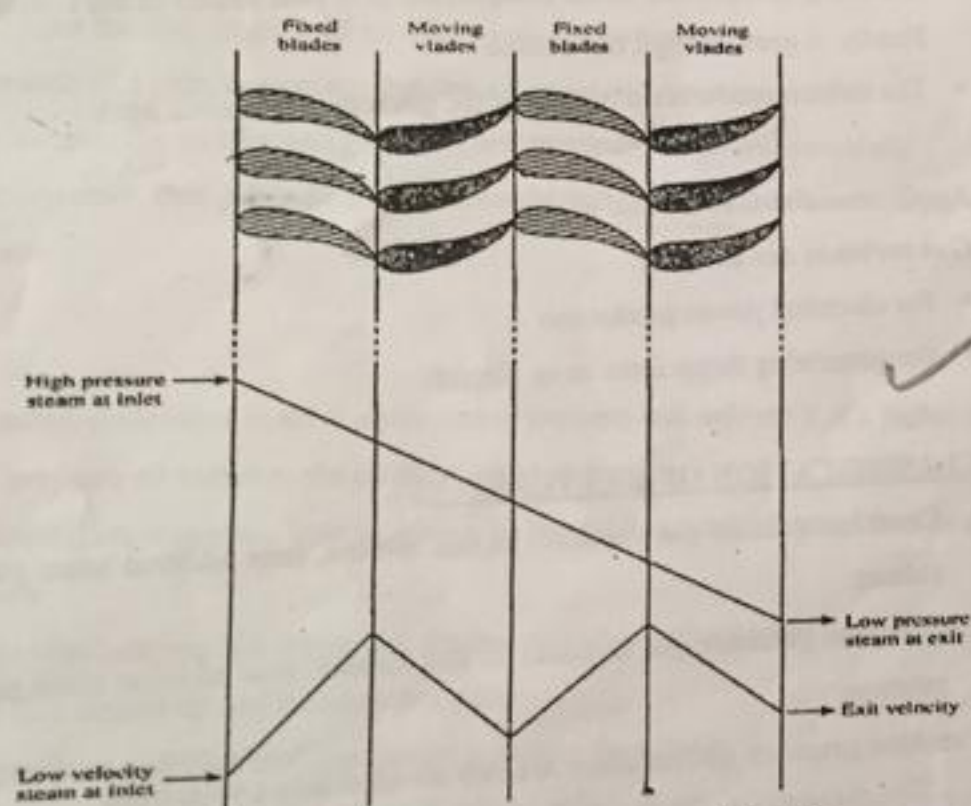


Fig c: P-V Diagram of Reaction Steam Turbine

- Fig b shows forces acting on the moving blade
- Fig c shows Pressure-Velocity diagram of Reaction steam turbine

- In operation, high pressure steam from the boiler enters to the first stage of fixed blades
- In the first stage of fixed blades, steam expands **partially** and hence pressure decreases and velocity increases
- The high velocity steam leaving the first stage of fixed blades is made to flow over the first stage of moving blades
- In the moving blades, steam undergoes changes in direction and velocity and hence changes in momentum and this gives Impulse force on the moving blade
- And also when the steam leaves the moving blade, a **Reactive force** is developed
- Thus, the net force acting on the moving blade is the sum of **Impulse force if Incoming steam and Reactive force of the outing steam**
- The reduction of pressure is further continues in the next stage of moving blades and fixed blades

Blades shape of Impulse and Reaction steam turbine

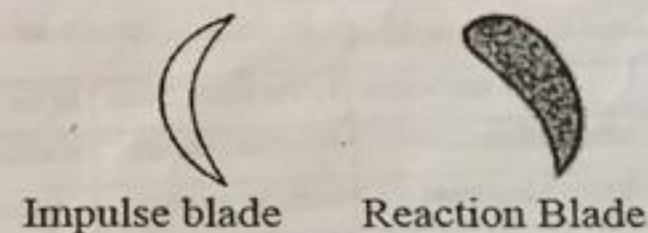
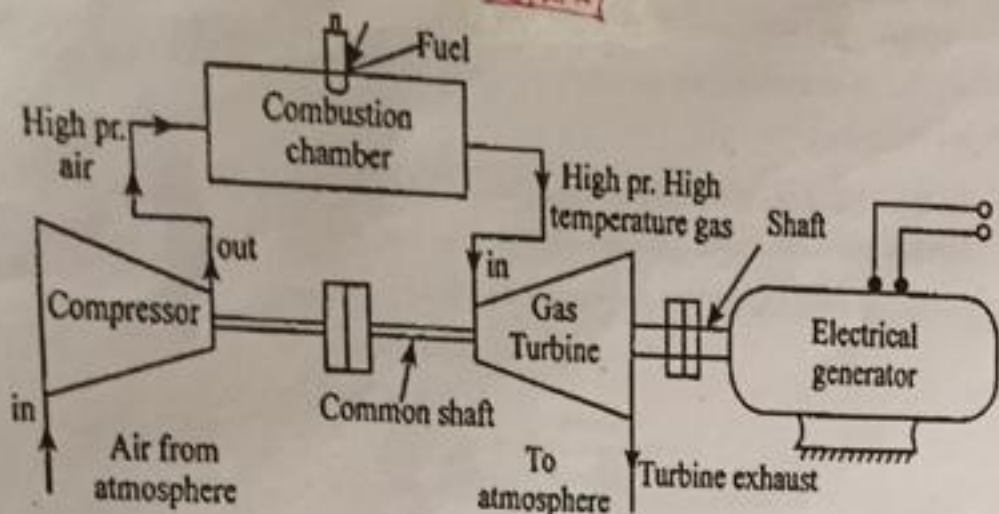


Fig: Blades shape of Impulse and Reaction turbine

Blade shapes of reaction turbines are designed in such a way that the steam flowing between the blades will be subjected to the *nozzle effect*. Hence, the pressure of the steam drops continuously as it flows over the blades causing, simultaneous increase in the velocity of the steam.

1. OPEN CYCLE GAS TURBINE:



- Fig shows the line diagram of an open cycle gas turbine
- It consists of a compressor, combustion chamber and a turbine
- The compressor, turbine and generator are coupled (connected) co-axially
- A compressor that takes in air at atmospheric pressure and delivers it at a higher pressure

Working:

- The compressor takes in air at atmospheric pressure and delivers it at a higher pressure
- High pressure air enters to the combustion chamber
- In combustion chamber, heat is added to the air by combustion of fuel at constant pressure
- Due to combustion, hot gases are formed and they mixed with compressed air and hence gets heated up and its mass increases
- High pressure and high temperature air enters to the turbine
- Expansion of hot gases takes place in the turbine and hence heat energy is converted into mechanical work
- The mechanical work is used to generate the electrical power
- The gases coming out from the turbine are discharged to atmosphere

2. CLOSED CYCLE GAS TURBINES:

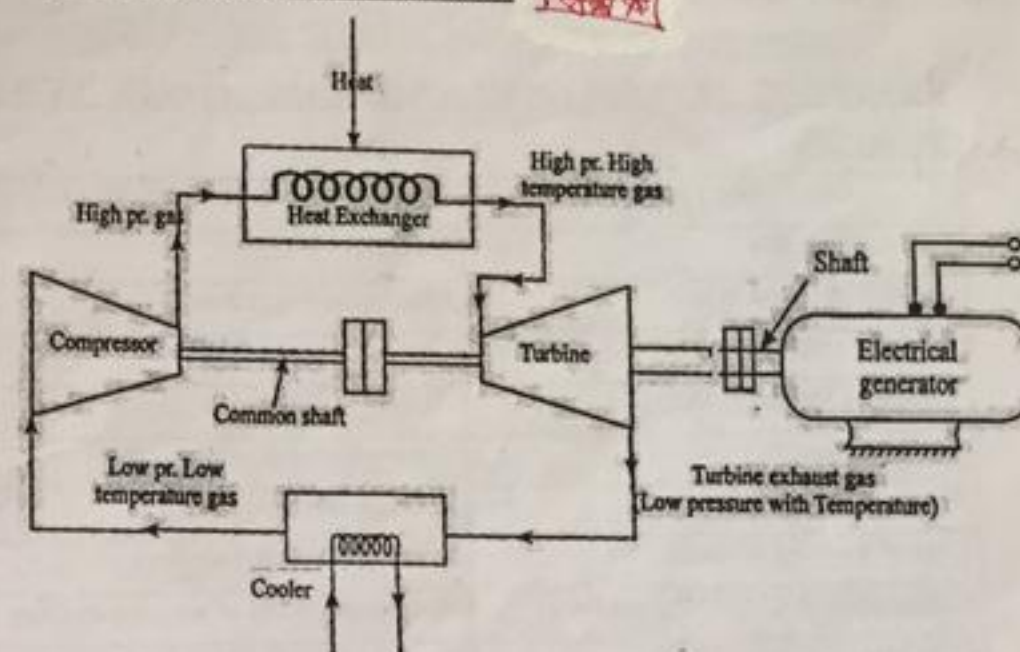


Fig: Closed cycle gas turbine

- Fig shows the line diagram of closed cycle gas turbine
- It consists of a compressor, heater, cooler and a turbine
- The compressor, turbine and generator are coupled (connected) co-axially

Working:

- The compressor, compresses the working medium (air or suitable gas) to high pressure and delivers it at a higher pressure
- High pressure gas enters to the heater chamber
- In heater, gas is heated to high temperature
- High pressure and high temperature air enters to the turbine
- Expansion of hot gases takes place in the turbine and hence heat energy is converted into mechanical work
- The mechanical work is used to generate the electrical power
- The gas coming out from the turbine is cooled in the cooler

WORKING OF IMPULSE WATER TURBINE OR PELTON WHEEL TURBINE

1. PELTON WHEEL TURBINE (IMPULSE WATER TURBINE):

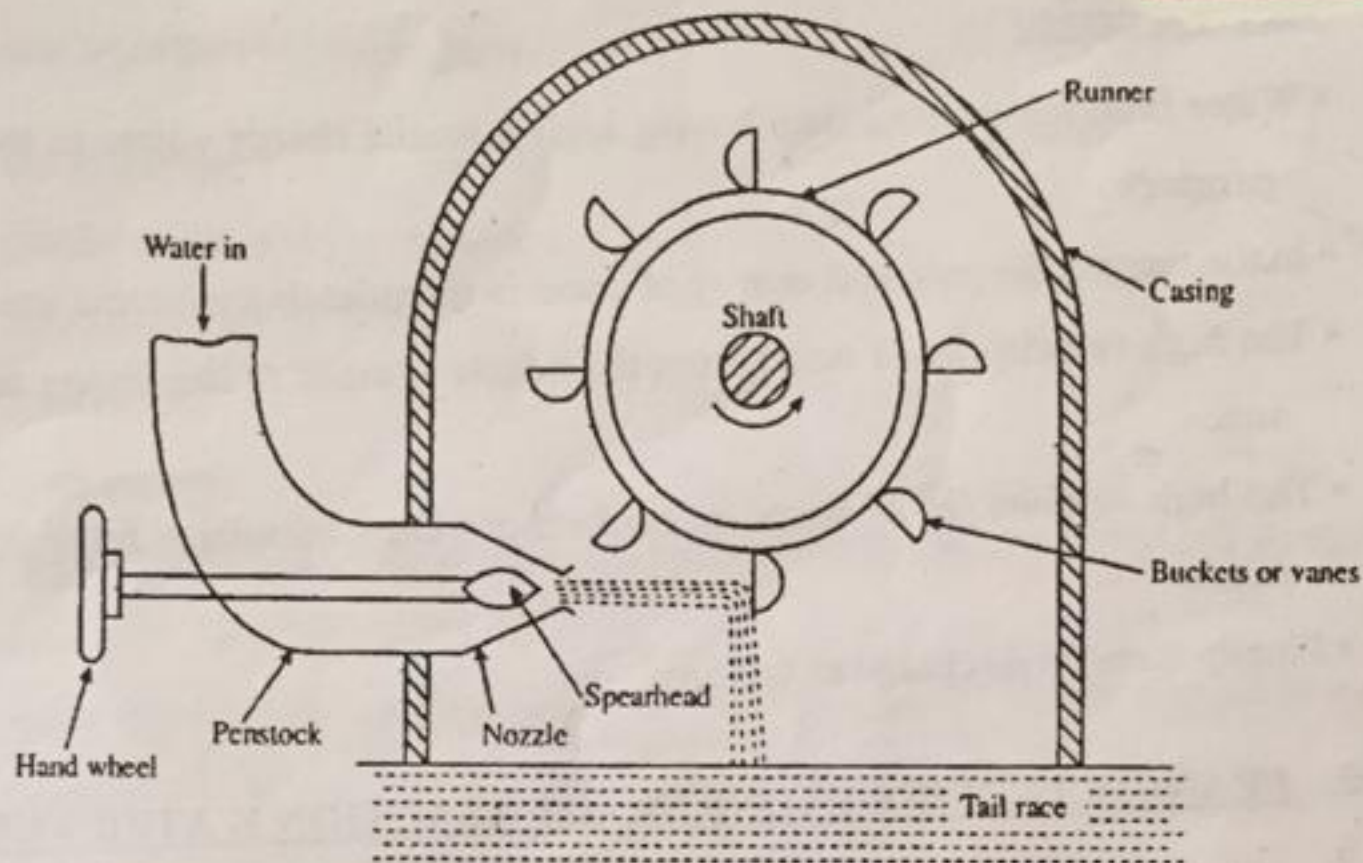


Fig : Pelton Wheel Turbine

- Fig shows Pelton wheel turbine
- Pelton wheel turbine is a tangential flow impulse turbine.
- It consists of following main parts
 - a. Nozzle with a spear head
 - b. Runner and buckets (vanes)
 - c. Casing

a. Nozzle with a spear head:

- Nozzle is attached to the penstock as shown in fig.
- Spear head is placed inside the nozzle.
- The flow of water through the nozzle is controlled by spear head
- The spear head is operated by a hand wheel

b. Runner and buckets(vanes):

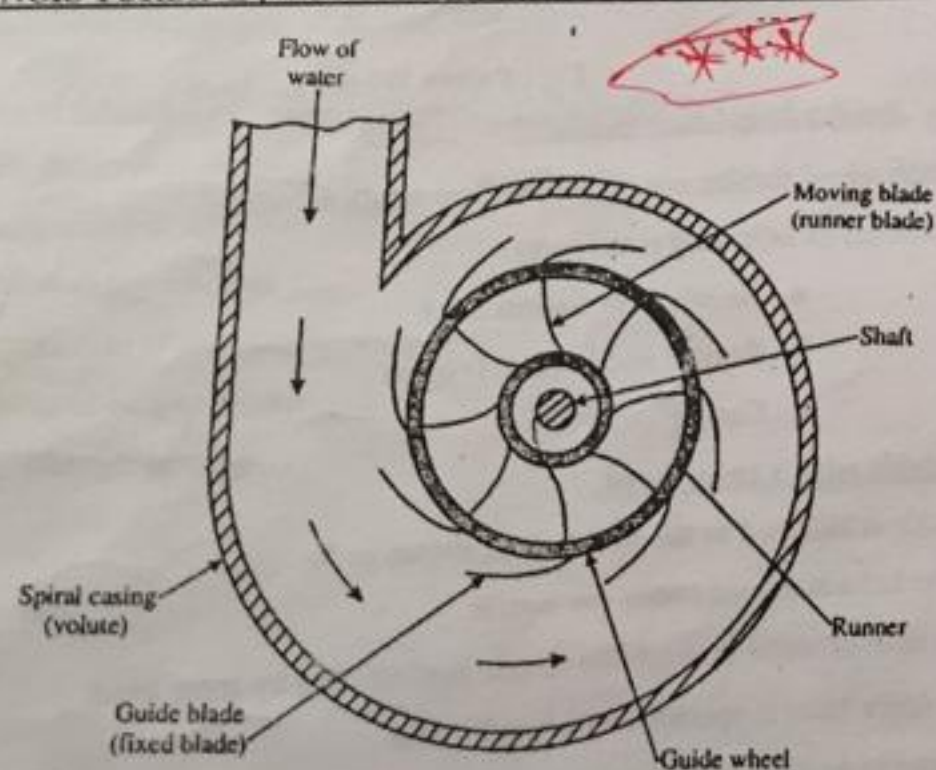
- Runner is a revolving wheel.
- Buckets or vanes or curved blades are fixed around the circumference of the runner.
- Runner is connected to shaft. Shaft is connected to electrical generator.

c. Casing:

- Pelton wheel turbine is placed inside.
- The function of casing is to prevent the splashing of the water and it also helps to discharge the water into the tailrace.

Working principle:

- Water from reservoir or dam having high potential energy enters to the nozzle through penstock.
- In the nozzle, the potential energy of water is completely converted into kinetic energy.
- The high velocity jet of water from the nozzle is made to impinge on the buckets or vanes.
- The high velocity jet of water on buckets produces impulsive force which rotates the runner.
- Finally water is discharge to tailrace.

2. FRANCIS TURBINE [WORKING OF REACTION WATER TURBINE]**Fig a : Francis Turbine**

- Fig a shows Francis turbine
- Francis turbine is a mixed flow reaction turbine

It consists of following main parts

- Scroll casing
- Guide wheel and guide vanes
- Runner
- Draft tube

a. Scroll casing:

- It is a closed passage whose cross-sectional area gradually decreases along the flow direction.
- The main function of scroll casing is to provide uniform distribution of water around the runner.

b. Guide wheel and guide vanes:

- Guide wheel consists of guide vanes on its periphery.
- Guide vanes direct the water onto moving vanes.

c. Runner:

- Runner is a revolving wheel.
- A series of curved blades are fixed around the runner
- Runner is connected to shaft.
- Shaft is connected to electrical generator.

d. Draft tube:

- It is a tube or pipe having gradually increasing area.
- It is used to discharge the water from turbine exit to tailrace at low pressure.

Working principle:

- Water from reservoir enters to scroll casing through the penstock.
- In the scroll casing, a part of the potential energy of water is converted into kinetic energy.
- Water from scroll casing is then enters to the guide vanes.

- Guide vanes direct the water on to the moving vanes or blades
- When water flows over the moving blades the remaining part of the potential energy is converted into kinetic energy. Thus velocity of water increases.
- This high velocity water leaves the blades and giving a reaction force which is responsible for runner rotation. Or when the water leaves the blade at high velocity a reaction force is set up and this force rotates the runner.

3. KAPLAN TURBINE (REACTION WATER TURBINE)

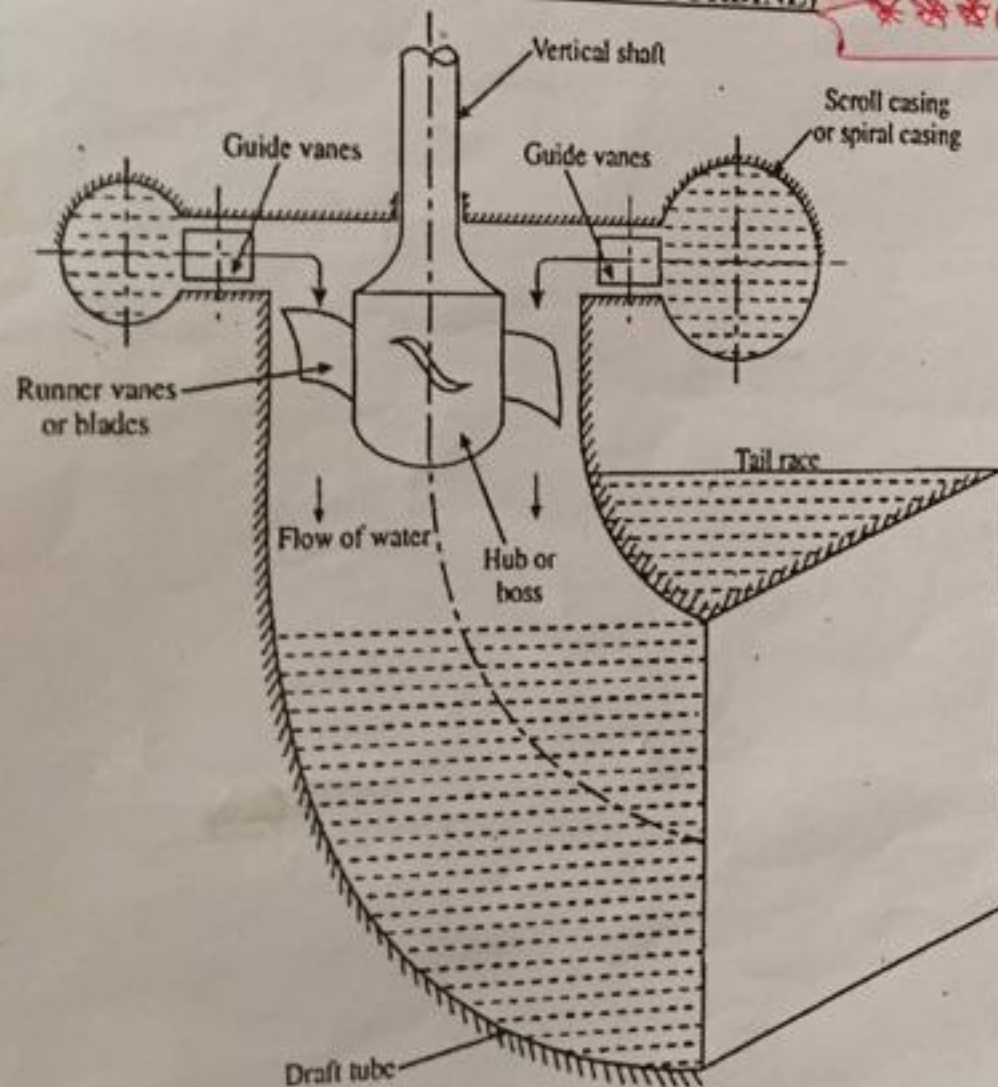


Fig: Kaplan Turbines

- Fig shows Kaplan turbine
- Kaplan turbine is a axial flow reaction turbine
- It is similar to Francis turbine except the runner and draft tube.

It consists of following main parts

- Scroll casing
- Guide wheel and guide vanes
- Runner or Hub or Boss
- Draft tube

a. Scroll casing:

- It is a closed passage whose cross-sectional area gradually decreases along the flow direction.
- The main function of scroll casing is to provide uniform distribution of water around the runner.

b. Guide wheel and guide vanes:

- Guide wheel consists of guide vanes on its periphery
- Guide vanes direct the water onto moving vanes.

c. Runner or Hub or Boss:

- The runner (Boss or Hub) resembles with the propeller of the ship, hence it is called as Propeller turbine
- The runner or Hub is fixed to a vertical shaft.
- A series of curved blades or moving blades are fixed around the hub
- The moving blades are in the range of 3 to 8
- The shaft is connected to electrical generator.

d. Draft tube:

- It is a tube or pipe having gradually increasing area.
- It is used to discharge the water from turbine exit to tailrace at low pressure.

- Working principle:**
- Water from reservoir enters to scroll casing through the penstock.
 - In the scroll casing, a part of the potential energy of water is converted into kinetic energy.
 - Water from scroll casing enters to guide vanes.
 - Guide vanes direct the water on to moving vanes.
 - When water flows over the moving blades the remaining part of the potential energy is converted into kinetic energy. Thus velocity of water increases.
 - This high velocity water leaves the blades and giving a reaction force which is responsible for runner rotation. Or when the water leaves the blade at high velocity a reaction force is set up and this force rotates the runner.

Applications: Kaplan turbines are widely used throughout the world for electrical power production. They suitable for low head hydro sites and for high flow rate conditions

Pelton wheel turbine	Francis turbine	Kaplan turbine or Propeller turbine
Impulse turbine	Reaction turbine	Reaction turbine
Tangential flow turbine	Mixed flow turbine	Axial flow turbine
High head turbine	Medium head turbine	Low head turbine

Some of the locations in India where turbines are installed are:

Pelton Turbine

- Sharavathi [Karnataka]
- Koyana [Maharashtra]
- Ballinala [Kerala]
- Puthur [Kerala]
- Joginder Nagar [Himachal Pradesh]

Francis Turbine

- Shila Sandakata [Karnataka]
- Bhakra [Punjab]
- Chandri Nagar [Rajasthan]
- Rihand [U.P.]

Kaplan Turbine

- Tungbhadra [Karnataka]
- Nizam Nagar [Andhra Pradesh]
- Kolhapur [Maharashtra]
- Orange & Koka [Punjab]

DIFFERENCE BETWEEN PELTON WHEEL, FRANCIS AND KAPLAN TURBINE

Pelton wheel turbine	Francis turbine	Kaplan turbine or Propeller turbine
It was designed by Allen Pelton	It was designed by James Francis	It was designed by Kaplan Victor
Impulse turbine	Reaction turbine	Reaction turbine
Works on principle of Newton's 2 nd Law	Works on principle of Newton's 3 rd Law	Works on principle of Newton's 2 nd Law
Tangential flow turbine	Mixed flow turbine	Axial flow turbine
High head turbine	Medium head turbine	Low head turbine
No draft tube	draft tube present	draft tube present
Nozzle is present	Scroll casing is present	Scroll casing is present
More number of blades present on the runner	More number of blades present on the runner	Less number of blades present on the runner
It requires small quantity of water	It requires medium quantity of water	It requires large quantity of water

DIFFERENCE BETWEEN IMPULSE AND REACTION WATER TURBINES

Impulse water turbine	Reaction water turbine
Works on principle of Newton's 2 nd Law	Works on principle of Newton's 3 rd Law
The blades have symmetrical profile	The blades have asymmetrical profile
Nozzle is present	No Nozzle is present
Draft tube is not necessary	Draft tube is necessary
High head turbine	Medium and low head turbine
It requires small quantity of water	It requires medium and large quantity of water
Ex: Pelton wheel	Ex: Francis and Kaplan Turbines